

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 437171483

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.473514 d
Epoch (t0) 1543.8678 BTJD
Duration 0.0840 d
Depth 4.947 % (49,472 ppm)
SNR 922.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

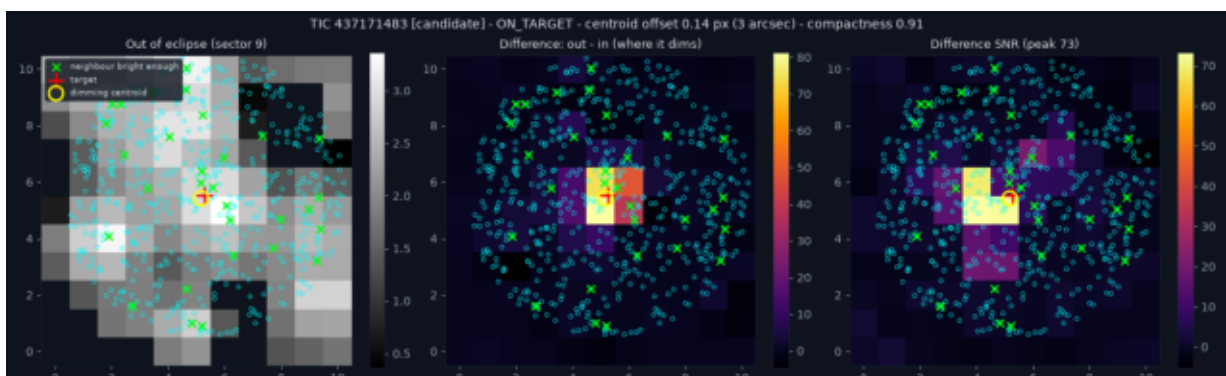
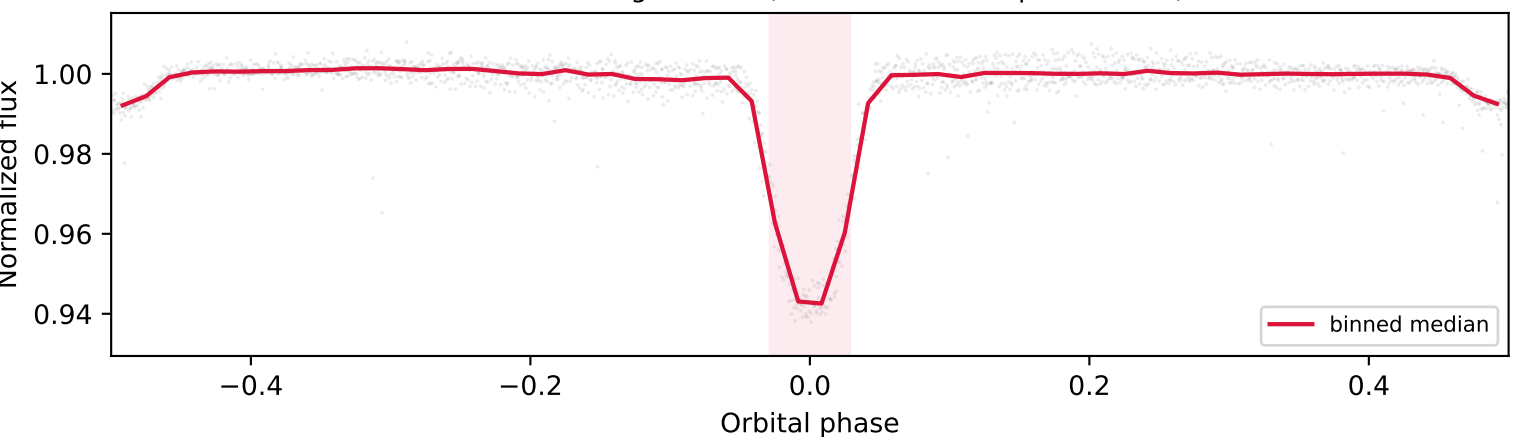
Eclipses observed 30 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 173.5 (gate ≥ 7.0)
Odd/even z-score 1.65 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 127.7 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 73.37 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.14 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 437171484
0.55 px away, dTmag +3.05 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 411457490

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 2.432352 d (ALIASED, see below)
-> corrected period: 4.864705 d
Epoch (t0) 1596.8966 BTJD
Duration 0.1700 d
Depth 5.531 % (55,313 ppm)
SNR 431.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

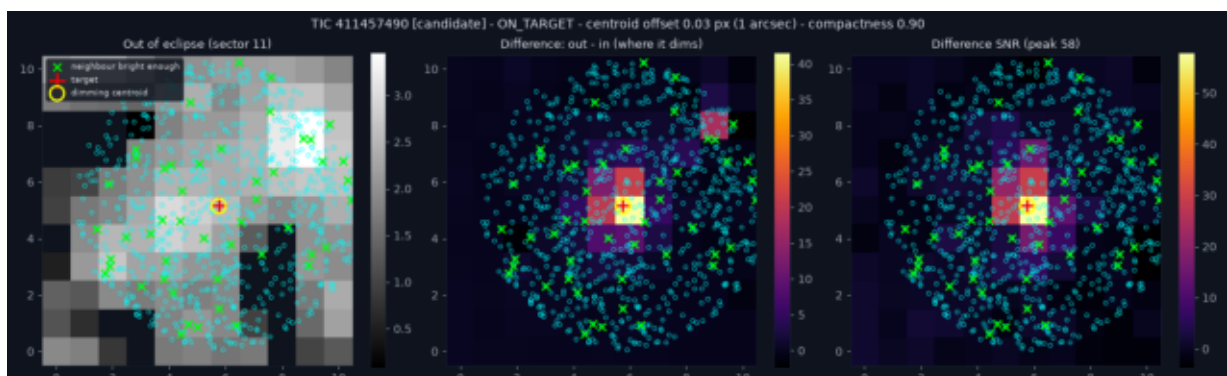
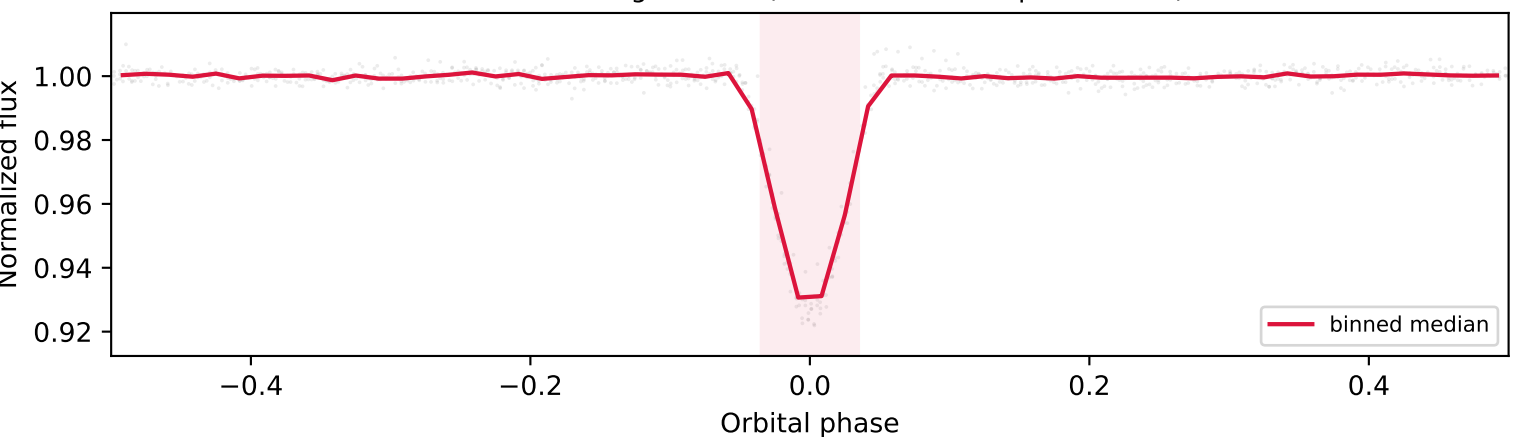
Eclipses observed 12 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 243.6 (gate ≥ 7.0)
Odd/even z-score 4.70 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 231.3 sigma t0 refined by 0.0100 phase
Difference-image peak SNR 58.00 (gate ≥ 5.0) Compactness 0.90 (gate ≥ 0.50)
Dimming centroid offset 0.03 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 411457450
0.94 px away, dTmag +2.68 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.70 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.864705 d, in any write-up.

TIC 295373509

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.421872 d
Epoch (t0) 1654.9396 BTJD
Duration 0.1700 d
Depth 8.026 % (80,259 ppm)
SNR 246.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

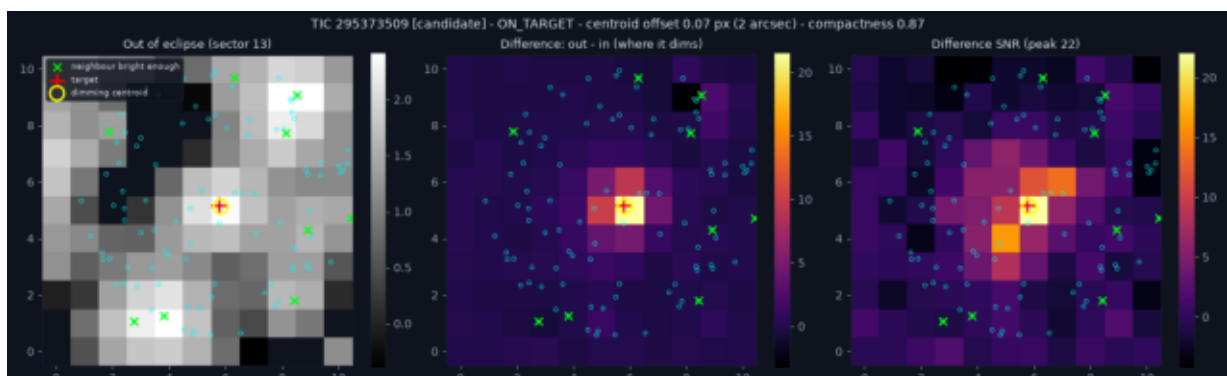
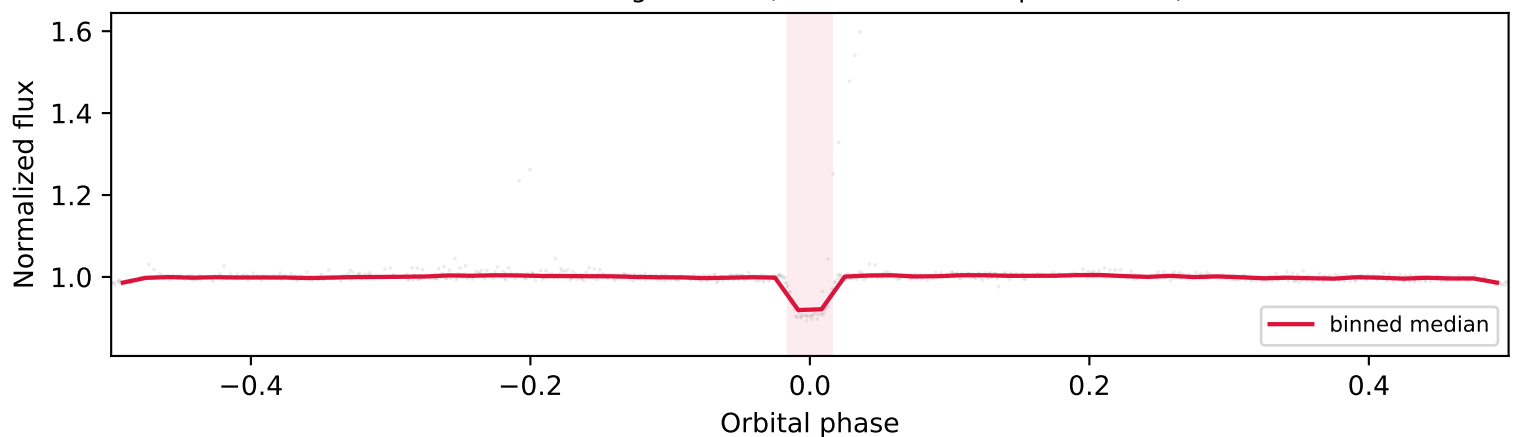
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.01 (gate ≥ 0.50)
Robust depth sigma 86.4 (gate ≥ 7.0)
Odd/even z-score 0.02 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 83.4 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 22.12 (gate ≥ 5.0) Compactness 0.87 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 295373523
3.19 px away, dTmag +2.13 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 374158907

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	14.656290 d (ALIASED, see below)	VSX	no match
-> corrected period:	29.312581 d	ASAS-SN	no match
Epoch (t0)	1551.5656 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1200 d	Gaia variable classification	no match
Depth	4.974 % (49,740 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	178.6	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	4 (gate >= 3)		
Robust depth ratio	1.14 (gate >= 0.50)		
Robust depth sigma	125.0 (gate >= 7.0)		
Odd/even z-score	14.90 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase 0.26, 7.4 sigma		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

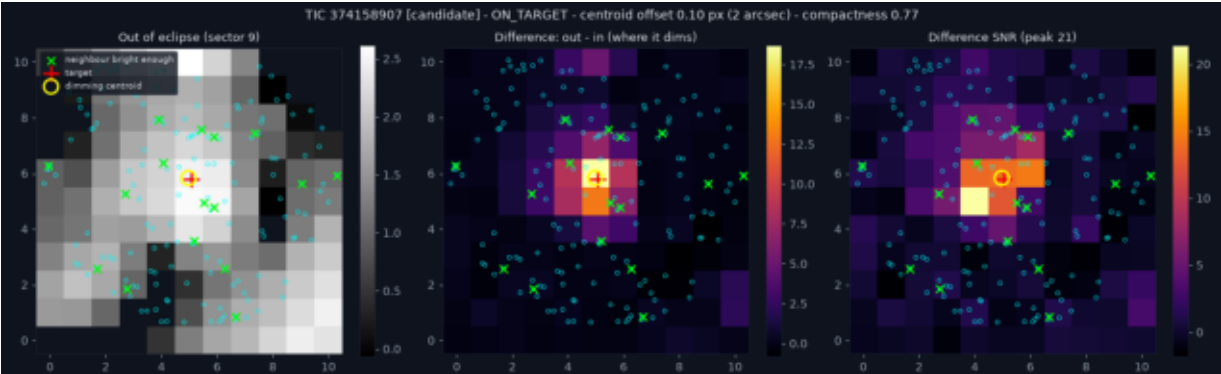
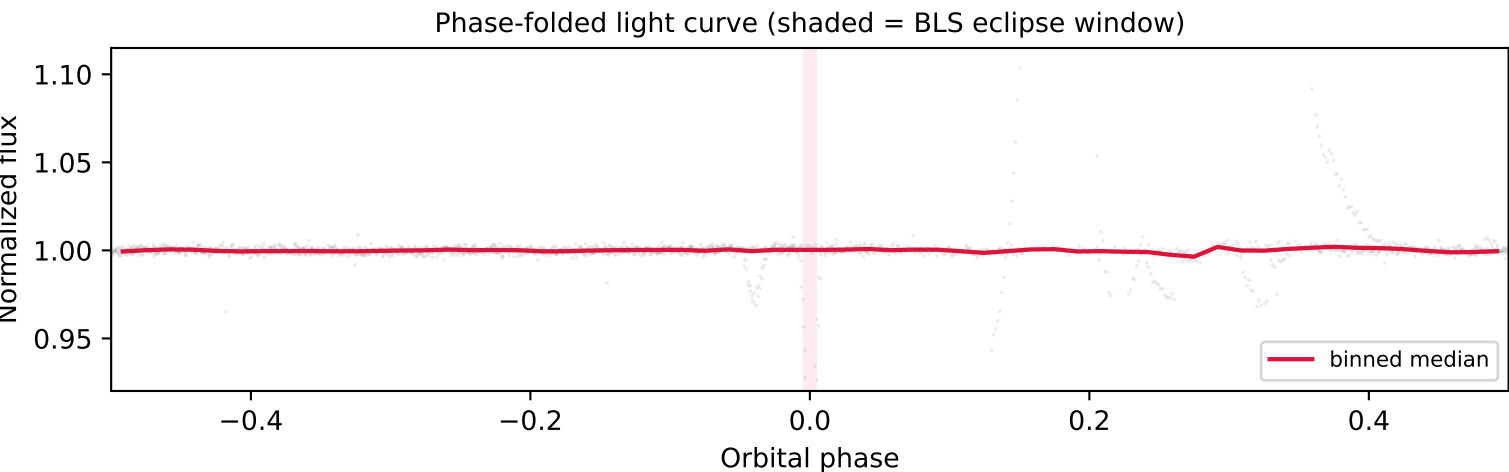
Sector 9 light-curve dip in this sector: 82.6 sigma t0 refined by 0.0000 phase

Difference-image peak SNR 21.40 (gate >= 5.0) Compactness 0.77 (gate >= 0.50)

Dimming centroid offset 0.10 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 374158897

1.05 px away, dTmag -0.74 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 14.90 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 29.312581 d, in any write-up.

TIC 348884775

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 6.731631 d
Epoch (t0) 1519.7050 BTJD
Duration 0.1700 d
Depth 14.751 % (147,512 ppm)
SNR 111.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

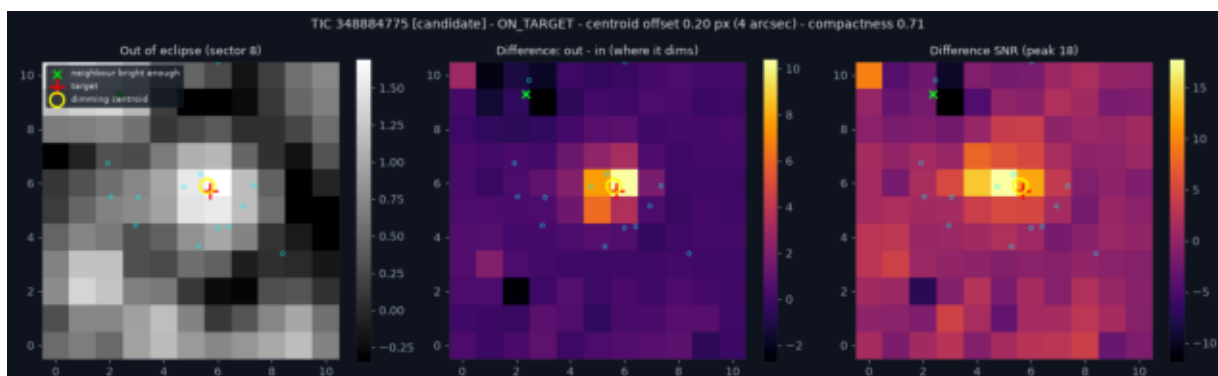
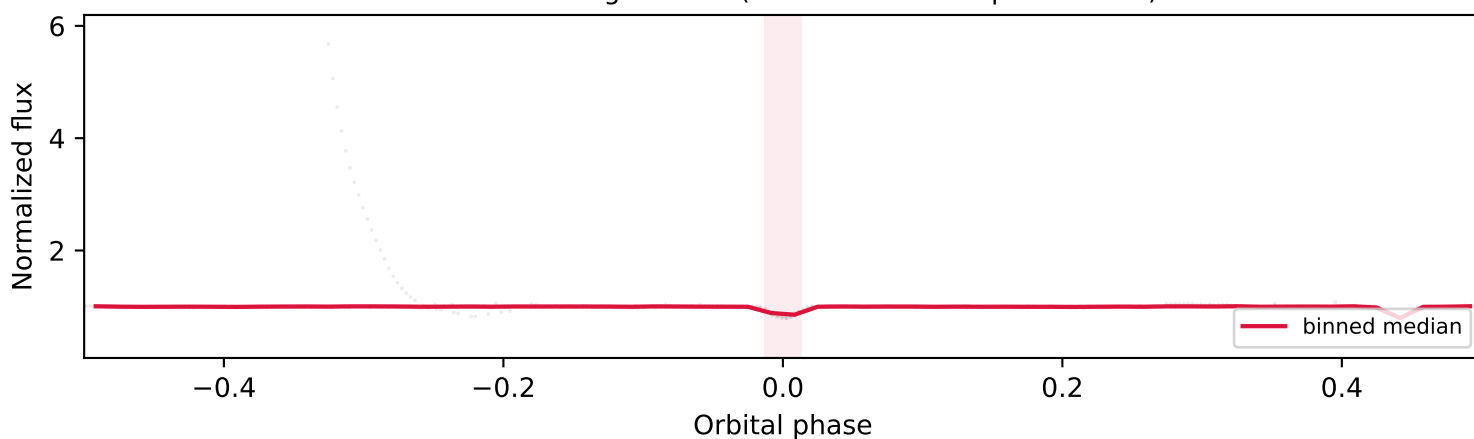
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 80.0 (gate ≥ 7.0)
Odd/even z-score 0.39 (consistent)
Secondary eclipse detected, phase 0.43, 7.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 8 light-curve dip in this sector: 66.5 sigma t0 refined by 0.4425 phase
Difference-image peak SNR 17.82 (gate ≥ 5.0) Compactness 0.71 (gate ≥ 0.50)
Dimming centroid offset 0.20 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 348884765
4.70 px away, dTmag +0.97 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 70762350

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.791085 d
Epoch (t0) 1355.3548 BTJD
Duration 0.1200 d
Depth 7.147 % (71,472 ppm)
SNR 66.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

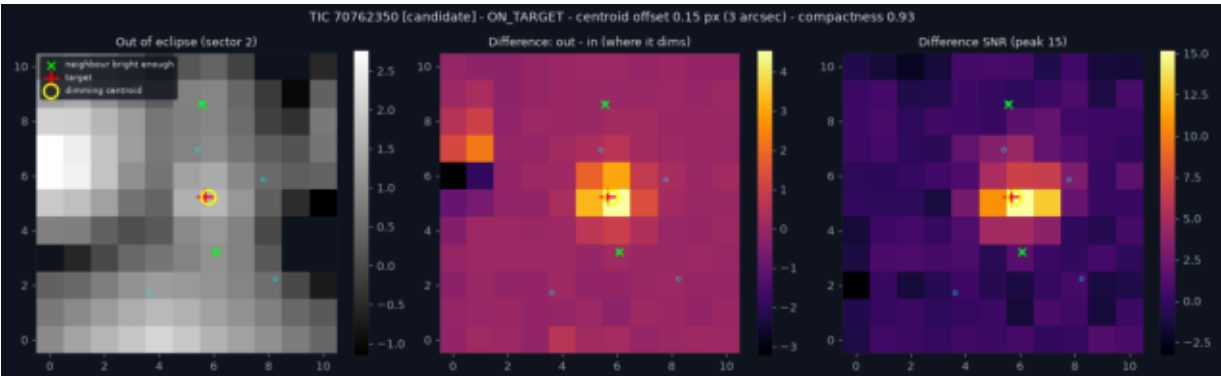
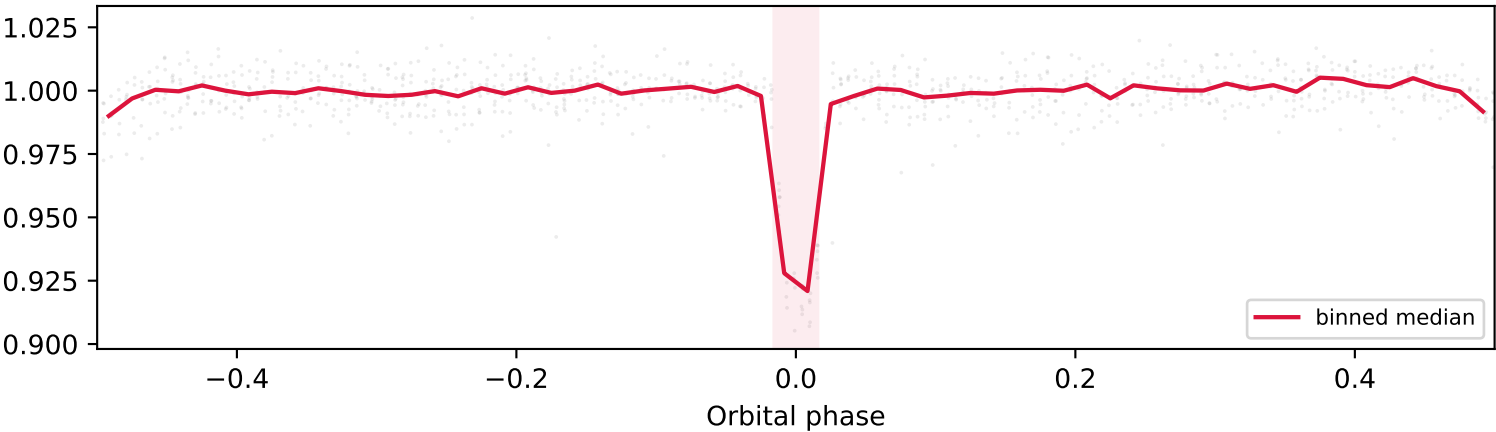
Eclipses observed 7 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 51.1 (gate ≥ 7.0)
Odd/even z-score 0.70 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 2 light-curve dip in this sector: 43.1 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 15.20 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 70762352
2.01 px away, dTmag +2.45 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.107957 d	VSX	no match
Epoch (t0)	3480.6522 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	0.231 % (2,310 ppm)	Gaia variable classification	no match
SNR	12.5	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	16 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	0.91 (gate >= 0.50)		
Robust depth sigma	9.9 (gate >= 7.0)		
Odd/even z-score	0.18 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

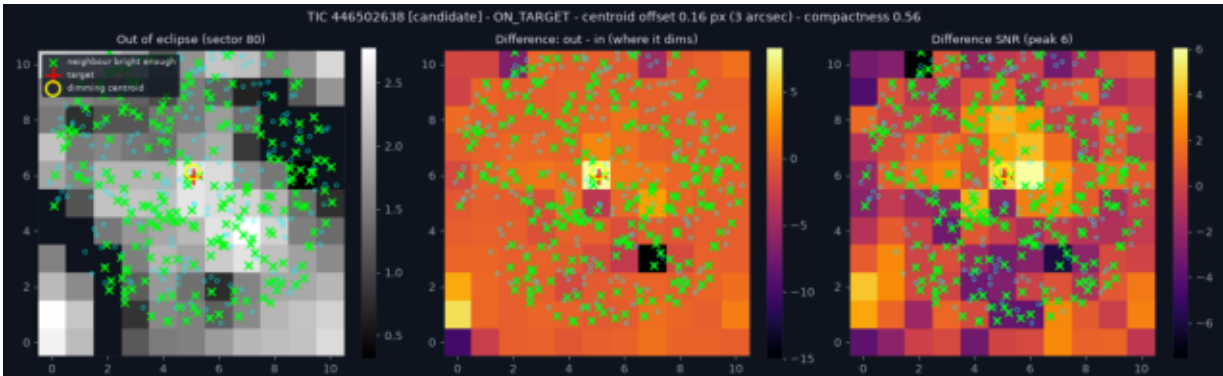
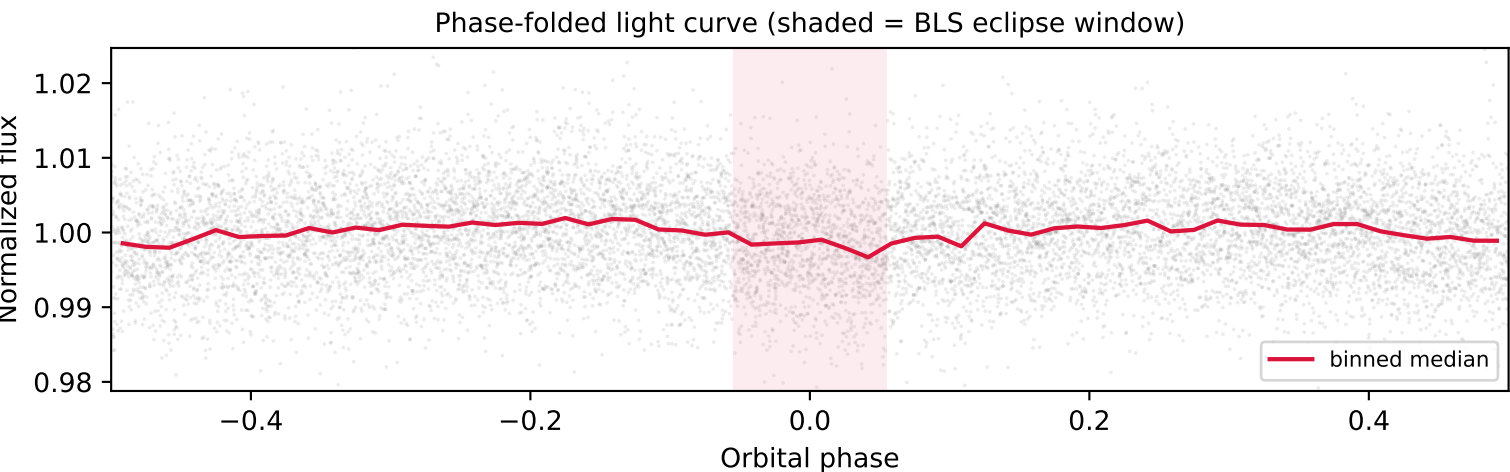
Sector 80 light-curve dip in this sector: 11.1 sigma t0 refined by 0.0300 phase

Difference-image peak SNR 6.06 (gate >= 5.0) Compactness 0.56 (gate >= 0.50)

Dimming centroid offset 0.16 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1748977177

0.25 px away, dTmag +6.56 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.090 px (dimming centroid 0.158 px from target vs 0.247 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.